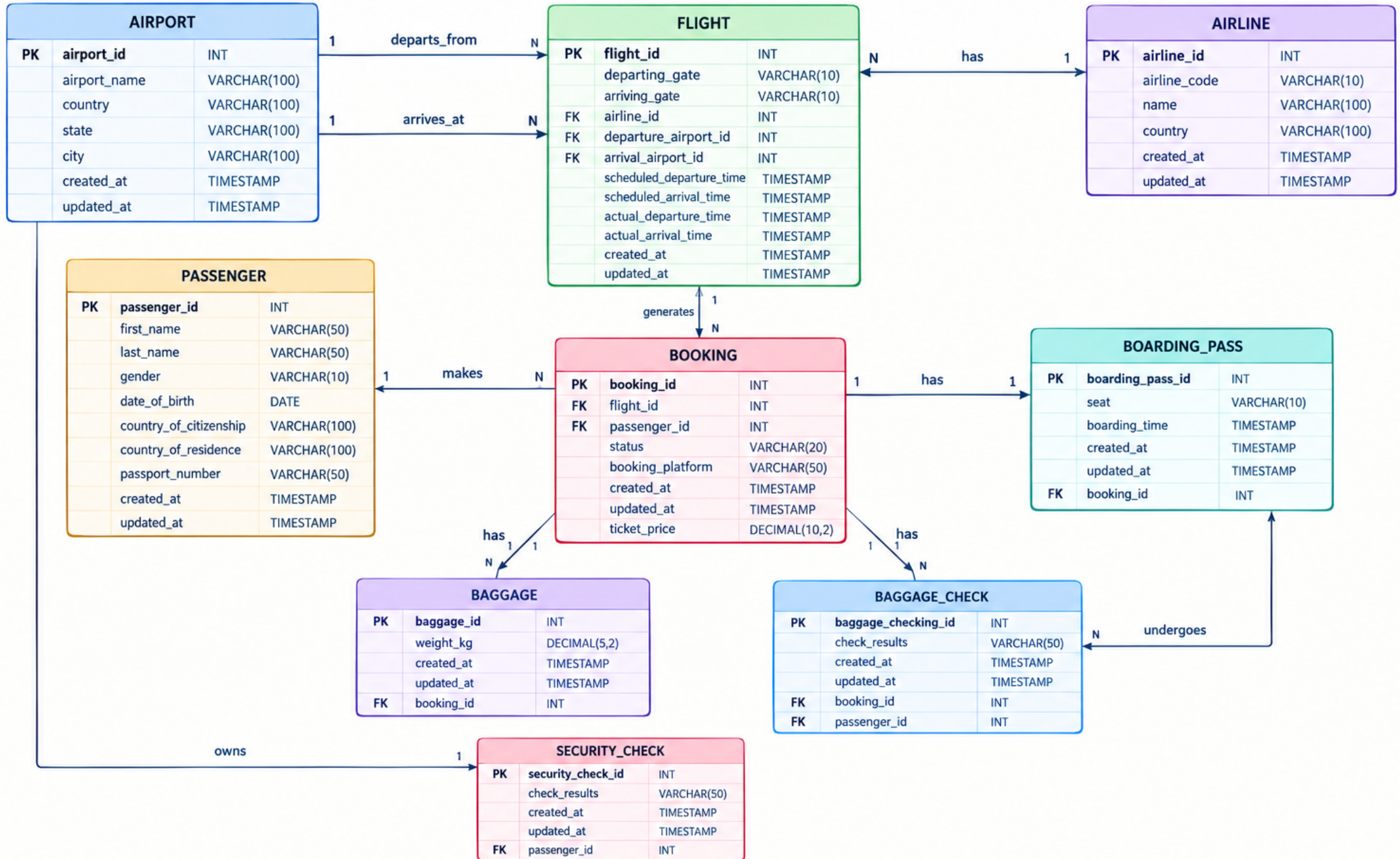


International Airport – ER Diagram

Lab 1: Entity-Relationship Diagram



Explanation

1. Airport

Stores airport_id, airport_name, country, state, city, created_at and updated_at. airport_id is the primary key.

2. Airline

Stores airline_id, airline_code, name, country, created_at and updated_at. One airline can have many flights (1:N).

3. Flight

Stores flight, gate, scheduled/actual time and timestamp information. airline_id, departure_airport_id and arrival_airport_id are foreign keys. A flight has one airline and one departure and arrival airport.

4. Booking

Stores booking_id, flight_id, passenger_id, status, platform, timestamps and ticket price. It connects passengers to flights. One flight and one passenger can each have many bookings.

5. Passenger

Stores passenger identity/profile information including name, gender, date of birth, citizenship, residence, passport number and timestamps.

6. Boarding Pass

Stores boarding_pass_id, seat, boarding_time, timestamps and booking_id. This diagram represents Booking → Boarding Pass as 1:1.

7. Baggage

Stores baggage_id, weight_kg, timestamps and booking_id. One booking can have many baggage records (1:N).

8. Baggage Check

Stores baggage checking id, check results, timestamps, booking_id and passenger_id. It is associated with both a booking and a passenger.

9. Security Check

Stores security_check_id, check results, timestamps and passenger_id. It is associated with a passenger.

Legend

PK = Primary Key. **FK** = Foreign Key. **1:N** = one-to-many. **1:1** = one-to-one.

Source note: The diagram is based on the nine entities explicitly described in the supplied laboratory work. The text also mentions accounts, booking legs, aircraft information and frequent flyer programs, but does not provide their attributes as entities, so they are not added here.